Implementation and experimentation with Dynamic Staging

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Compile-Time vs. Run-Time Work

Programmers often write code in a clear, general style, even when parts of it could be computed during compilation.

Original program

```mls
fun dot(xs, ys, i) =
  if xs.length == i then 0
  else xs.(i) * ys.(i)
    + dot(xs, ys, i + 1)

fun dotWith3(v) =
  dot([1, 0, 2], v, 0)
```

can actually be written as

```mls
fun dotWith3(v) = v(0)
+ 2 * v(2)
```

Compile-Time vs. Run-Time Work

The recursion, the pattern matching, the multiplication are all known at compile time. The residual program contains only the work that genuinely depends on the runtime input v.

Goal: let the programmer keep the abstract version, and have the compiler peel away the static layer automatically.

Multi-Stage Programming

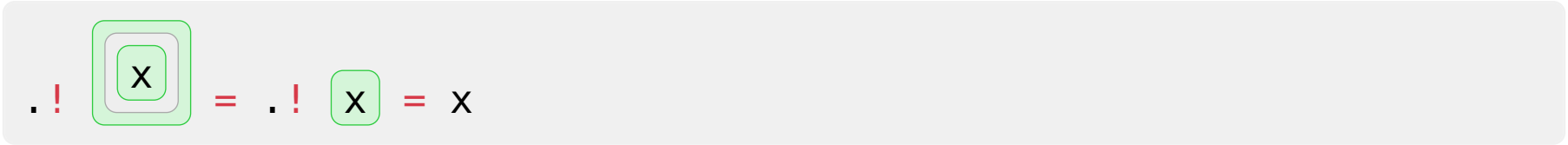
Well-known optimization technique [1]

```
1  fun power(n, x) = if n is
2    0 then .<1>.
3    else .<.~x * .~(power(n - 1, x))>.
4  fun power2 = .! .<(x => .~(power(2, .<x>.)))>.
```

In here, green annotates code to be executed in the next stage, and gray executed in the current stage.

Multi-Stage Programming

Rule for .!: evaluate until we the whole code fragment is in the next stage, then move it to the current stage.



During evaluation, we able to pre-compute certain parts of the code, reducing runtime.

Multi-Stage Programming

```
1  x => power(2, x)                                                                    mls
2  x => if 2 is 0 then 1 else x * power(2 - 1, x)
3  x => x * power(1, x)
4  x => x * if 1 is 0 then 1 else x * power(1 - 1, x)
5  x => x * x * if 0 is 0 then 1 else x * power(0 - 1, x)
6  x => x * x * 1
7  fun power2 = x => x * x * 1
```

Class specialization

```
1  class B(val x) with
2    fun f() = ...
3  // ...
4  if x is
5    1 then new D1(1)
6    2 then new D2(2)
7
8  x.f()
```

```
…                    …
8   if x is                    mls
9     D1 then x.D1_f1()
10    D2 then x.D2_f1()
11    D3 then x.D3_f1()
```

Traditional Multi-Stage Programming tracks types.

Even if `x` has a known class shape, we cannot remove matching arms.

Specialization is done through typing, so we know that `x : B`, but we do not know which specific derived class of `B` is used.

Class specialization

```mls
data class Foo(x, y) extends Bar(x+1, y+1)

if new Foo(1, 2) is
  Bar(x, y) then ...
```

Under single inheritance, matching arms runs into problems. [2]

(Class splitting)

```mls
class Foo$1$2 extends Bar
class Bar$2$3
```

Approach in High Level

Start with an ordinary module:

```mls
module M with
  fun dot(xs, ys, i) =
    if xs.length == i then 0
    else xs.(i) * ys.(i) + dot(xs, ys, i + 1)
  fun dotWith3(v) = dot([1, 0, 2], v, 0)
```

Approach in High Level

Users can mark a module as `staged` which turns the module into a **code generator**. When we execute the program, it emits a new residual module instead of executing the program.

```mls
staged module M with
  fun dot(xs, ys, i) =
    if xs.length == i then 0
    else xs.(i) * ys.(i) + dot(xs, ys, i + 1)
  fun dotWith3(v) = dot([1, 0, 2], v, 0)
```

Approach in High Level

The output of executing the program is

```mls
module M with
  fun dotWith3(v) = v.(0) + 2 * v.(2)
  ... with more specialized functions ...
```

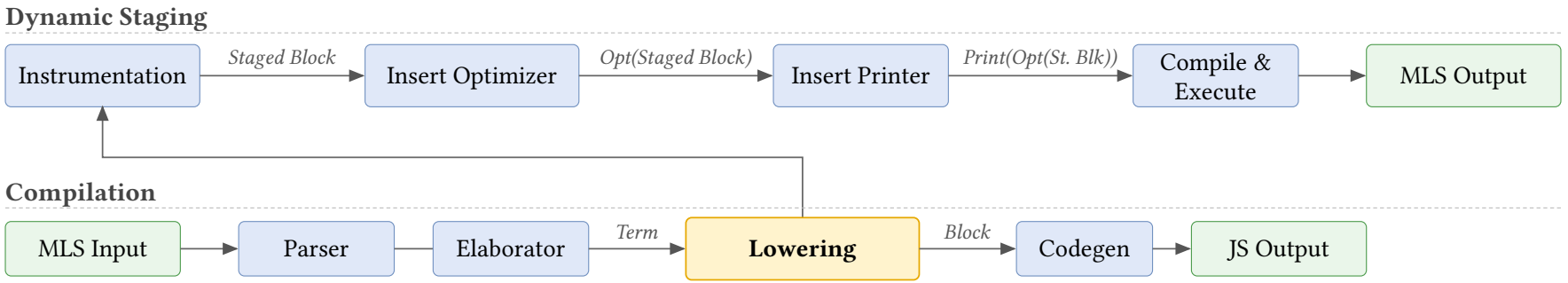
which the user can import.

Hence, our approach is a combination of **metaprogramming** (code generator) and **specialization** (generation of specialized functions).

MLscript Compiler

- Lexer/Parser: converts source code into AST
- Elaborator/Resolver: Deal with references
- **Lowering**: converts AST to a simplified format
- Codegen: Turn AST into executable code (JavaScript).

Dynamic Staging Recipe

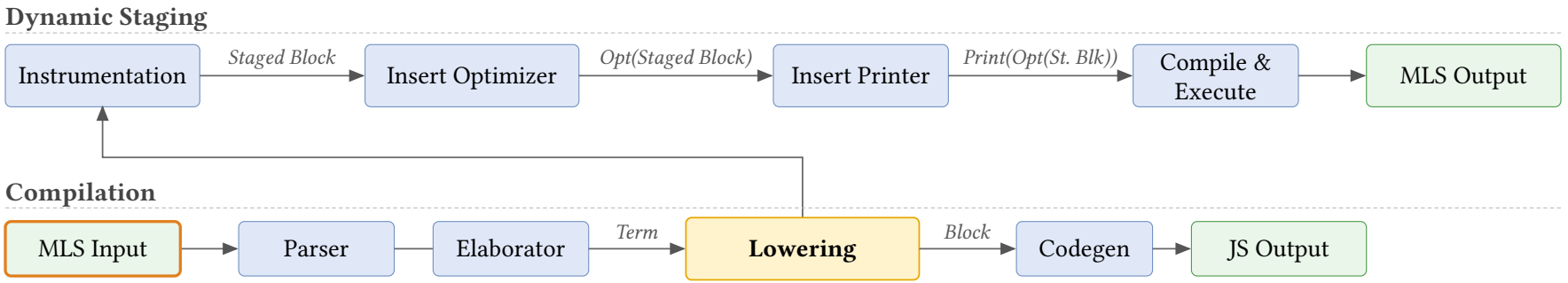


Two phases: Instrumentation / Insert Optimizer

Lowering Pass

We implement staging a transformation from Scala Block ⇒ Scala Block.

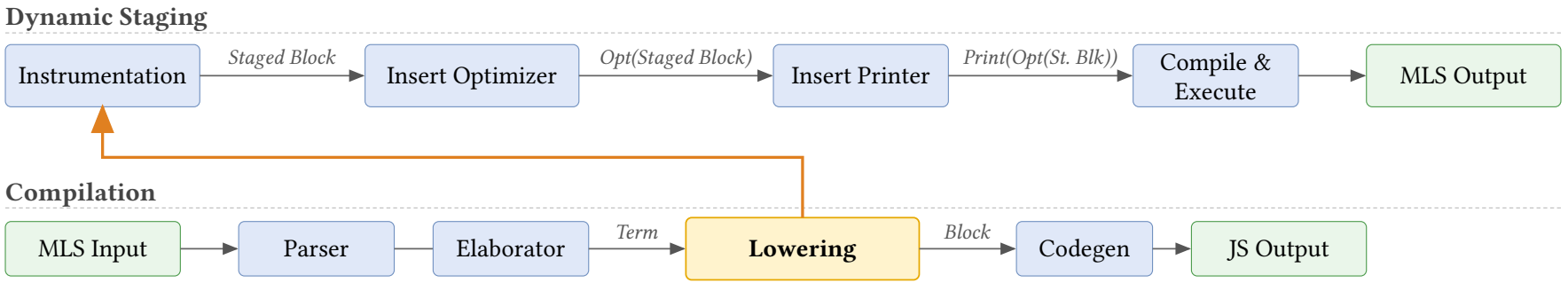
Dynamic Staging Recipe



Source (MLscript)

```mls
if C(0) is
  Int  then "Int"
  C(n) then "C(" + n + ")"
  else "Unknown"
```

Dynamic Staging Recipe

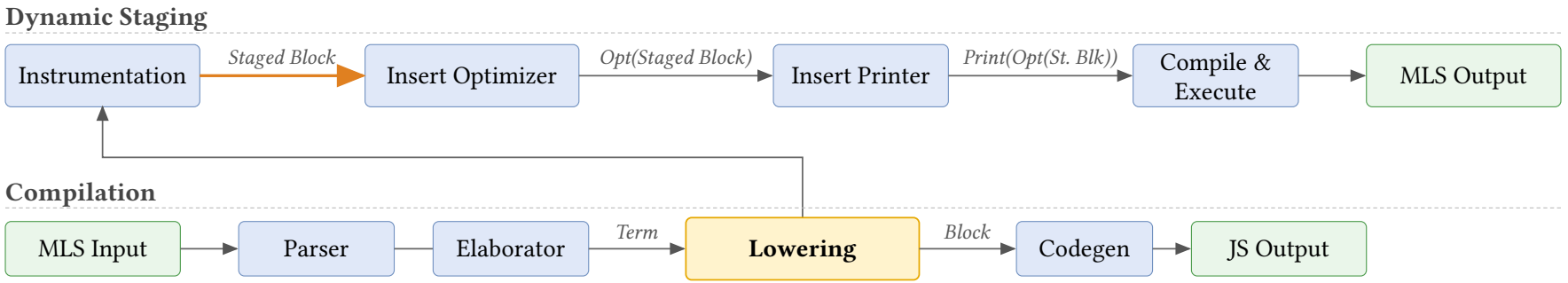


Block

Scala

```
Scoped([scrut, n, arg, tmp],
  Assign(scrut, Call(C, [0])),
  Match(Ref(scrut), [
    Arm(Cls(Int), Ret("Int")),
    Arm(Cls(C),
      Assign(arg, Select(scrut, "n")),
      Assign(n,   Ref(arg)),
      Assign(tmp, Call("+", ["C(", n])),
      Ret(Call("+", [tmp, ")"]))
  ], Ret("Unknown")))
```

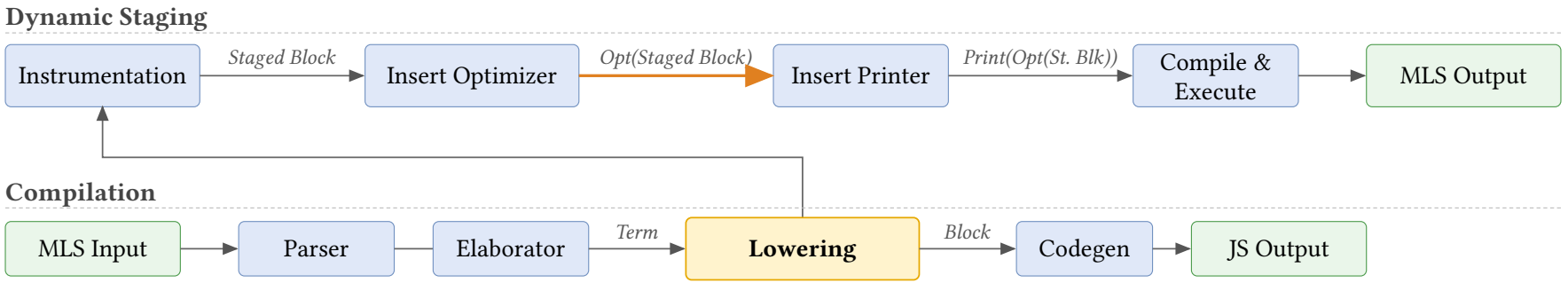
Dynamic Staging Recipe



Staged Block

```mls
Scoped([scrut, n, arg, tmp],
  Assign(scrut, Call(C, [0])),
  Match(Ref(scrut), [
    Arm(Cls(Int), Ret("Int")),
    Arm(Cls(C),
      Assign(arg, Select(scrut, "n")),
      Assign(n,   Ref(arg)),
      Assign(tmp, Call("+", ["C(", n])),
      Ret(Call("+", [tmp, ")"]))
  ], Ret("Unknown")))
```

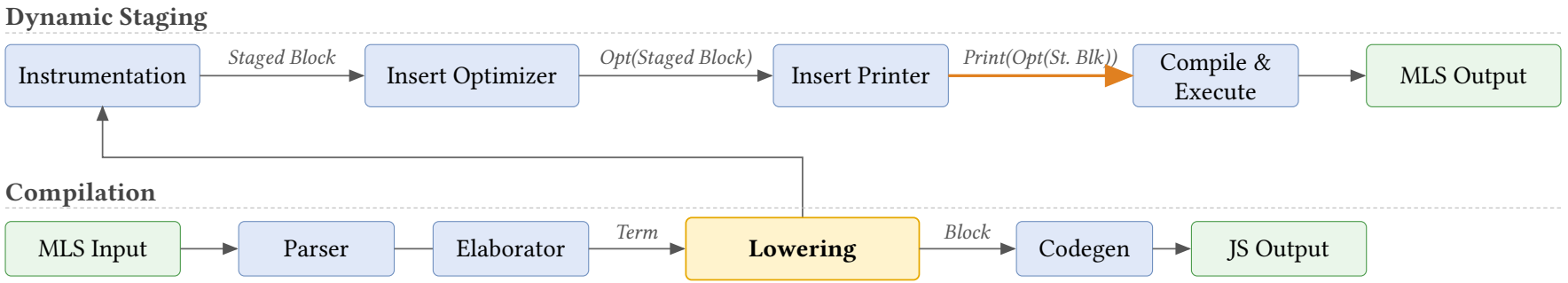
Dynamic Staging Recipe



Opt(Staged Block)

```
prop(                                                    mls
  Scoped([scrut, n, arg, tmp],
    Assign(scrut, Call(C, [0])),
    Match(Ref(scrut), [
      Arm(Cls(Int), Ret("Int")),
      Arm(Cls(C),
        Assign(arg, Select(scrut, "n")),
        Assign(n,   Ref(arg)),
        Assign(tmp, Call("+", ["C(", n])),
        Ret(Call("+", [tmp, ")"]))
    ], Ret("Unknown")))
)
```

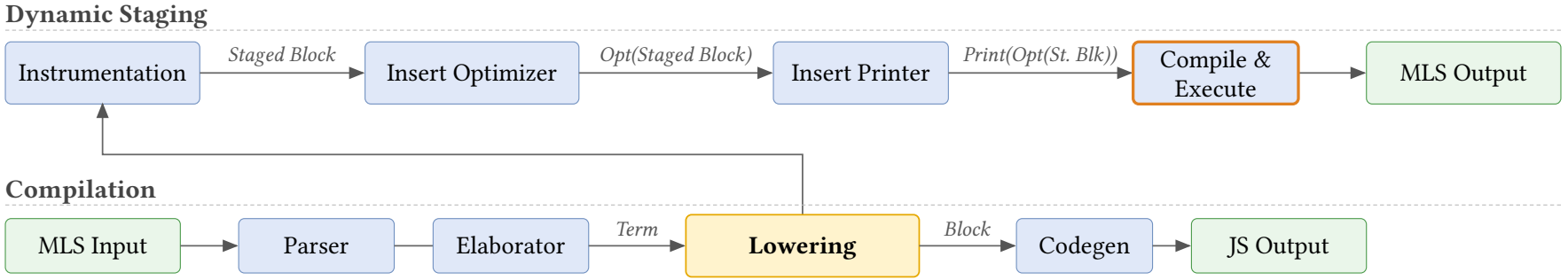
Dynamic Staging Recipe



Print(Opt(Staged Block))

```
print(                                                    mls
  prop(
    Scoped([scrut, n, arg, tmp],
      Assign(scrut, Call(C, [0])),
      Match(Ref(scrut), [
        Arm(Cls(Int), Ret("Int")),
        Arm(Cls(C),
          Assign(arg, Select(scrut, "n")),
          Assign(n,   Ref(arg)),
          Assign(tmp, Call("+", ["C(", n])),
          Ret(Call("+", [tmp, ")"]))
      ], Ret("Unknown")))
  )
)
```

Dynamic Staging Recipe

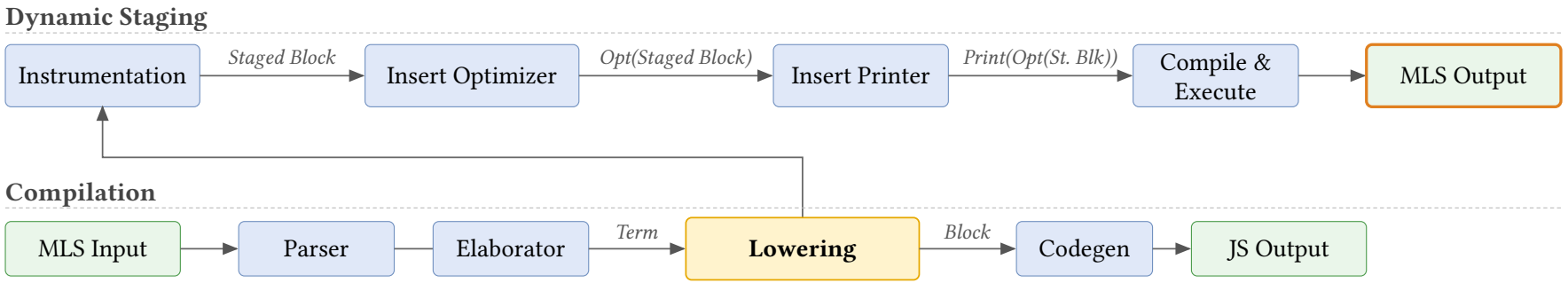


Compile & Execute

This execution is the **first stage** of metaprogramming. It entails the execution of

- the **optimizer** which outputs the **optimized staged block**,
- and the **printer** which prints the new MLscript program from the optimized staged block.

Dynamic Staging Recipe



Generated MLscript

```mls
"C(0)"
```

Staging Block

```scala
case class Return(res, implct) extends Block
case class Match(scrut, arms, dflt, rest) extends Block
case class Assign(lhs, rhs, rest) extends Block
```

Scala

```mls
class Block with
  constructor
    Return(res, implct)
    Match(scrut, arms, dflt, rest)
    Assign(lhs, rhs, rest)
```

mls

For any Scala Block data, we can recreate the same structure within MLscript which we called **Staged Block**.

Staging Block

```
fun pow(x, n) = if n is 0 then 1 else x * pow(x, n-1)
```

```scala
val pow = Symbol("pow"); val x = Symbol("x"); val n = Symbol("n")
val t1 = Symbol("tmp"); val t2 = Symbol("tmp")
val sub = Symbol("-"); val mul = Symbol("*")
FunDefn(pow, [x, n], Scoped(HashSet(t1, t2), Match(Ref(n),
  [[ Lit(0), Return(Lit(1)) ]],
  Assign(t1, Call(sub, [n, Lit(1)]),
    Assign(t2, Call(pow, [x, t1]),
      Return(Call(mul, [x, t2]))
    )
  ), End())
))
```

As long as we have the corresponding constructor, we can copy over the structure to the Staged Block.

Staging Block

```
fun pow(x, n) = if n is 0 then 1 else x * pow(x, n-1)
```

```mls
let pow = Symbol("pow"); let x = Symbol("x"); let n = Symbol("n")
let t1 = Symbol("tmp"); let t2 = Symbol("tmp")
let sub = Symbol("-"); let mul = Symbol("*")
FunDefn(pow, [x, n], Scoped([t1, t2], Match(Ref(n),
  [[ Lit(0), Return(Lit(1)) ]],
  Assign(t1, Call(sub, [n, Lit(1)]),
    Assign(t2, Call(pow, [x, t1]),
      Return(Call(mul, [x, t2]))
    )
  ), End())
))
```

As long as we have the corresponding constructor, we can copy over the structure to the Staged Block.

Staging Block

We perform structural induction to stage each type of Scala Block.

`Value.Lit(lit)`

```
ValueLit(lit)                    mls
```

Example: `1` $\mapsto$ `ValueLit(1)`

`Symbol(name)`

```
Symbol(name)                     mls
```

Example:

- `x` $\mapsto$ `Symbol("x")`
- `c` $\mapsto$ ...?

This is not enough for staging symbols. We'll revisit this case later on.

Staging Block

We perform structural induction to stage each type of Scala Block.

`Value.Lit(lit)`

```
ValueLit(lit)                                    mls
```

Example: `1` ↦ `ValueLit(1)`

`Value.Ref(sym)`

```
let l = sym                                      mls
ValueRef(l)
```

Example: `x` ↦ `ValueRef(Symbol("x"))`

`Select(qual, name)`

```
let q = qual                                     mls
let n = name
Select(q, n)
```

Example: `x.p` ↦ `Select(ValueRef(Symbol("x")), Symbol("p"))`

Staging Block

```
Tuple([x0, x1, ...])
```

```
let s0 = x0                                                    mls
let s1 = x1
                                    ...
Tuple([x0, x1, ...])
```

The other Scala Block cases are similar, staging the parameters of the Scala Block by induction and recreating the corresponding Staged Block counterpart.

Staging Symbols

- We need more information about the Symbol in the original stage

  Add reference to current value in the symbol

  `C.x`

```mls
let CSym = ClassSymbol("C", C)
Select(ValueRef(CSym), Symbol("x"))
```

  Add redirection within current stage to allow access for next stage

```mls
staged class A with
  fun f() = M.f()
  val redirect_M = M
```

Staging Symbols

- Uniqueness of Symbols

  We want Symbols for the same object in the previous stage to be unique in the current stage across functions and compilations.

For local symbols, we can maintain a map during staging to reuse a staged symbol.

```
x + x
```

```mls
let x = Symbol("x")
// let x1 = Symbol("x")
Call(ValueRef(Symbol("+")), [[ValueRef(x), ValueRef(x)]])
```

Staging Symbols

- Uniqueness of Symbols

  We want Symbols for the same object in the previous stage to be unique in the current stage across functions and compilations.

For class and module symbols, we need to cache and use first instance of a symbol within the staged code.

```
M.f()
```

```
1  let MSym = symbolMap.check(ModuleSymbol("M", M))          mls
2  Call(Select(ValueRef(MSym), Symbol("f")), [[]])
```

Instrumentation

Insert some auxiliary helper variables/functions to the module for shape propagation.

mls

```
staged module Math with
  val funCache = new Map()
  val generatorMap = new Map([["pow", pow_gen]])
  fun pow_staged() = ...
  fun pow_gen(x, n) =
    specialize(funCache, "f", pow_staged, [x, n])
  fun propagate() =
    let dyn = ...
    pow_gen(dyn, dyn)
    sq_gen(dyn)
```

Block in BNF

Symbol x, y, z *Literal* ι

Function definition $\mathbf{fun}\ f(\overline{x_i}^n) = b$ *Plain class definition* $\mathbf{class}\ C\left(\overline{\mathbf{val}\ n}\right)$

Staged module definition $\mathbf{staged\ module}\ M\ \mathbf{with\ fun}\ f(\overline{x_i}^n) = b$

$$
\begin{array}{rrcl}
\textit{Path} & p & ::= & \iota \mid x \mid p.n \\
\textit{Result} & r & ::= & p \mid f(\overline{p}) \mid \mathbf{new}\ C(\overline{p}) \mid [\overline{p}] \\
\textit{Match pattern} & \pi & ::= & \iota \mid C \mid []^n \mid \_ \\
\textit{Match arm} & \kappa & ::= & \pi\ \mathbf{then}\ b \\
\textit{Non-tail block} & B & ::= & \varepsilon \mid \mathbf{if}\ p\ \mathbf{is}\ \overline{\kappa}; B \mid x = r; B \mid \mathbf{let}\ x; B \\
\textit{Block} & b & ::= & \mathbf{return}\ r \mid \mathbf{end} \mid B; b
\end{array}
$$

Shape Definition

A **Shape** captures what we statically know about a value at compile time.

$$\text{Shape } s ::= \iota \mid \mathbf{dyn} \mid [\overline{s}] \mid C(\overline{n : s}) \mid \bot \mid s \cup s$$

We write ⟦s⟧ to denote the shape of an expression, distinguishing it from actual values.

```
1  fun f(p, cond) =                                  mls
2    let a = 42                                      ⟦42⟧
3    let b = C(p, 2)                                 ⟦C(dyn, 2)⟧
4    let c = if cond then [1, 2] else C(1, 2)        ⟦[1,2] ∪ C(1,2)⟧
5    let d = if b is C then 0 else 1                 ⟦0⟧
```

The mechanism to calculate the shapes are called **Shape Propagation**.

Shape Definition

A Tiny Example

```mls
staged module M with
  fun add1(x) = x + 1
  fun test()  = add1(2)
```

Walking through `test()`:

1. Argument `x` has shape ⟦2⟧ (a literal).
2. Specialise `add1` with `x` ↦ ⟦2⟧ inside the body, `x` looks up ⟦2⟧ in the context.
3. `x + 1` folds: ⟦2⟧ + ⟦1⟧ → ⟦3⟧.
4. Cache the result as `add1_Lit2() = 3`; rewrite the call site `add1(2)` to `add1_Lit2()`.

Final shape of `test()`: ⟦3⟧. Body collapses to a constant.

Shape of Path ($\Gamma \vdash p \Rightarrow s$)

We evaluate paths p to their shapes s.

$$\frac{}{\Gamma \vdash \iota \Rightarrow [\![\iota]\!]}$$

Example

```
42                    mls
```

$\Gamma = \varepsilon$

$\varepsilon \vdash 42 \Rightarrow [\![42]\!]$

$$\frac{(p \mapsto s) \in \Gamma \quad s \neq \bot}{\Gamma \vdash p \Rightarrow s}$$

Example

```
x                     mls
```

$\Gamma = x \mapsto [\![C(n : [\![1]\!])]\!]$

$\Gamma \vdash x \Rightarrow [\![C(n : [\![1]\!])]\!]$

$$\frac{p \notin \mathrm{dom}(\Gamma) \quad \Gamma \vdash p \Rightarrow s}{\Gamma \vdash p.n \Rightarrow \mathrm{sel}(s, [\![n]\!])}$$

Example

```
x.n                   mls
```

$\Gamma = x \mapsto [\![C(n : [\![1]\!])]\!]$

$\Gamma \vdash x.n \Rightarrow [\![1]\!]$

$$p ::= \iota \mid x \mid p.n$$

Shape of Path ($\Gamma \vdash p \Rightarrow s$)

Shape Selection (`sel`)

sel computes the shape of a selection.

$$\mathrm{sel}(\llbracket C(\overline{n_i : s_i}) \rrbracket, \llbracket n_i \rrbracket) = s_i \qquad \mathrm{sel}(\mathbf{dyn}, \llbracket n_i \rrbracket) = \mathbf{dyn}$$

$$\mathrm{sel}(\llbracket [\overline{s_i}^{i \in 0..n}] \rrbracket, \llbracket i \rrbracket) = s_i \qquad \mathrm{sel}(s_1 \cup s_2, s) = \mathrm{sel}(s_1, s) \cup \mathrm{sel}(s_2, s)$$

$$\mathrm{sel}(s_1, s_2) = \mathbf{err}$$

Example

$$\begin{aligned}\mathrm{sel}(\llbracket [1,2] \rrbracket \cup \llbracket [2,3,4] \rrbracket, \llbracket 1 \rrbracket) &= \mathrm{sel}(\llbracket [1,2] \rrbracket, \llbracket 1 \rrbracket) \cup \mathrm{sel}(\llbracket [2,3,4] \rrbracket, \llbracket 1 \rrbracket) \\ &= \llbracket 2 \rrbracket \cup \llbracket 3 \rrbracket\end{aligned}$$

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

We optimize block and track shapes simultaneously: r becomes r' with shape s.

$$\frac{\Gamma \vdash p \Rightarrow s}{\Gamma \vdash p \rightsquigarrow p \Rightarrow s}$$

Example

```
x                    mls
```

$\Gamma = x \mapsto [\![1]\!]$
$\Gamma \vdash x \rightsquigarrow x \Rightarrow [\![1]\!]$

$$\frac{\Gamma \vdash p_i \Rightarrow s_i}{\Gamma \vdash [\overline{p_i}^n] \rightsquigarrow [\overline{p_i}^n] \Rightarrow [\![[\overline{s_i}^n]]\!]}$$

Example

```
[x, y]               mls
```

$\Gamma = x \mapsto [\![1]\!], y \mapsto [\![2]\!]$
$\Gamma \vdash [x, y] \rightsquigarrow [x, y] \Rightarrow [\![[[\![1]\!], [\![2]\!]]]\!]$

$$\frac{\Gamma \vdash p_i \Rightarrow s_i}{\Gamma \vdash \mathbf{new}\ C(\overline{p_i}) \rightsquigarrow \mathbf{new}\ C(\overline{p_i}) \Rightarrow [\![C(\overline{n_i : s_i})]\!]}$$

Example

```
new Pair(x, y)       mls
```

$\Gamma = x \mapsto [\![1]\!], y \mapsto [\![2]\!]$
$\Gamma \vdash \mathbf{new}\ \mathrm{Pair}(x, y) \Rightarrow [\![\mathrm{Pair}(a : [\![1]\!], b : [\![2]\!])]\!]$

$$r ::= p \mid [\overline{p}] \mid \mathbf{new}\ C(\overline{p}) \mid f(\overline{p})$$

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

Shape of Result (sor) — Staged Application

When f is **staged**, we recursively specialise its body on the argument shapes.

$$\frac{f \quad \text{is staged} \quad \Gamma \vdash p_i \Rightarrow s_i \quad f_{\text{gen}}\,(\overline{p}, \overline{s}) = (r', s)}{\Gamma \vdash f(\overline{p_i}) \rightsquigarrow r' \Rightarrow s}$$

Example

```
staged module Staged with                                    mls
  fun pyth(x, y) = x * x + y * y
  fun test() = pyth(2, 3)
```

Specialise $\text{pyth}_{\text{gen}}\,([\![2]\!], [\![3]\!]) \rightarrow$ produces `pyth_Lit2_Lit3() = 13`, shape $[\![13]\!]$. The specialized function will be saved to the cache.

Conclusion: $\varepsilon \vdash \text{pyth}(2, 3) \rightsquigarrow 13 \Rightarrow [\![13]\!]$

$$r ::= p \mid [\overline{p}] \mid \textbf{new}\; C(\overline{p}) \mid f(\overline{p})$$

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

Shape of Result (sor) — Staged Application

After propagation, the call site `pyth(2, 3)` is rewritten to its cached variant.

Example

```mls
staged module Staged with
  fun pyth_Lit2_Lit3() = 13
  fun test() = pyth_Lit2_Lit3()
```

Cache

Function	Shapes	Symbol	Block	Returned Shape
`pyth`	$([\![2]\!], [\![3]\!])$	`pyth_Lit2_Lit3`	`return 13`	$[\![13]\!]$
`pyth`	$(\mathbf{dyn}, \mathbf{dyn})$	`pyth_Dyn_Dyn`	`return x*x+y*y`	$\mathbf{dyn}$

If f_{gen} is called with the same $(f, \overline{s})$ again, it looks up the existing entry in the cache instead of re-specialising.

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

Shape of Result (sor) — Non-Staged Application

When f is **non-staged**, we don't have its Staged Block. Two sub-cases.

$$\frac{f \text{ is non-staged} \quad \Gamma \vdash p_i \Rightarrow s_i \quad \forall i.\, \text{static}(s_i) \qquad f_{\text{imp}}\left(\overline{\text{valOf}(s)}\right) = (r, s)}{\Gamma \vdash f(\overline{p_i}) \rightsquigarrow r \Rightarrow s}$$

Example

```
NonStaged.sq(2)                                    mls
```

All args static; evaluate $\text{sq}(2) = 4$.
$\varepsilon \vdash \text{sq}(2) \rightsquigarrow 4 \Rightarrow [\![4]\!]$

$$\frac{f \text{ is non-staged} \quad \Gamma \vdash p_i \Rightarrow s_i \quad \exists i.\, \neg\, \text{static}(s_i)}{\Gamma \vdash f(\overline{p_i}) \rightsquigarrow f(\overline{p_i}) \Rightarrow \mathbf{dyn}}$$

Example

```
// where x is dynamic                              mls
NonStaged.sq(x)
```

$\Gamma = x \mapsto \mathbf{dyn}$; cannot evaluate, call remain unchanged.
$\Gamma \vdash \text{sq}(x) \rightsquigarrow \text{sq}(x) \Rightarrow \mathbf{dyn}$

$$r ::= p \mid [\overline{p}] \mid \mathbf{new}\, C(\overline{p}) \mid f(\overline{p})$$

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

Non-Staged Application Example

```
module NonStaged with                    mls
  fun sq(x) = x * x

staged module Staged with
  fun pyth(x, y) =
  NonStaged.sq(x) +
  NonStaged.sq(y)
  fun test() = pyth(2, 3)
```

After staging:

```
module Staged with                       mls
  fun pyth_Lit2_Lit3() = 13
  fun test() = 13
```

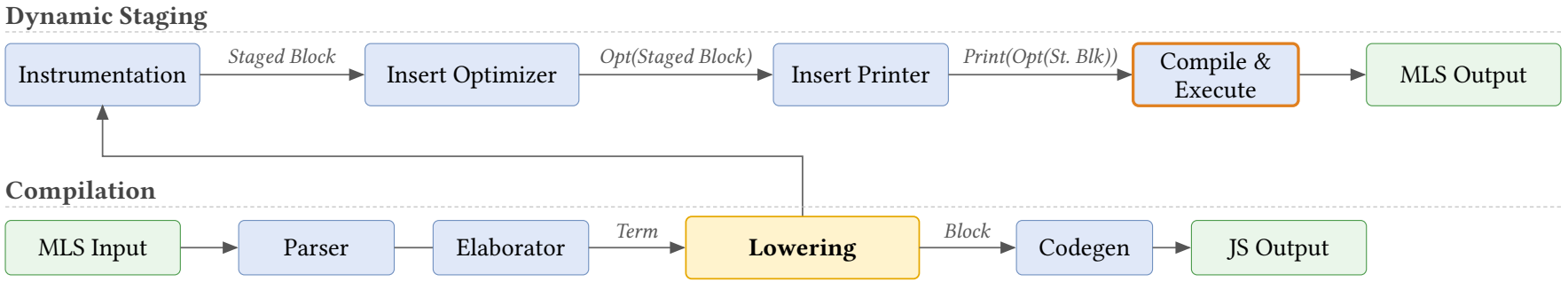
- Since every params of `sq(x)` is static, we will call `sq` to evaluate the result completely.
- No specialised functions are generated for the non-staged module `NonStaged`.

Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

How do we call the function `NonStaged.sq`?

Recall that during execution of the optimizer, we already compile the function `NonStaged.sq(x)` to JavaScript.

- Essentially we get the optimization for free in the static parameter case (at no expense of code size)



Shape of Result ($\Gamma \vdash r \rightsquigarrow r' \Rightarrow s$)

Static Shape

$$\text{static}([\![\iota]\!]) = \textbf{true} \qquad \text{static}([\![[\overline{s_i}^n]]\!]) = \bigwedge_i \text{static}(s_i)$$

$$\text{static}([\![C(\overline{n_i : s_i})]\!]) = \bigwedge_i \text{static}(s_i) \qquad \text{static}(s_1 \cup s_2) = \text{static}(s_1) \wedge \text{static}(s_2)$$

$$\text{static}(\textbf{dyn}) = \textbf{false} \qquad \text{static}(\bot) = \textbf{false}$$

Remark. $\text{static}(s)$ returns true iff no subshape of s contains **dyn**.

$$r ::= p \mid [\overline{p}] \mid \textbf{new}\ C(\overline{p}) \mid f(\overline{p})$$

Shape Propagation

Pattern Matching

Before giving the formal rules, let's give a simple trace of shape propagation for pattern matching.

```mls
class C(val n)
staged module Simple with
  fun f(x) =
    let y
    if x is C then y = x.n
    else y = 0
    y + 1
  fun test()     = f(C(2))
  fun test2(dyn) = f(C(dyn))
  fun test3()    = f(0)
```

Shape Propagation

Trace 1: `test() = f(C(2))`

Specialise `f` with $\texttt{x} \longmapsto [\![\texttt{C(n: 2)}]\!]$.

```
1  fun f(x) =                                  mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

$\texttt{x} \longmapsto [\![\texttt{C(n: 2)}]\!]$

Shape Propagation

Trace 1: `test() = f(C(2))`

Specialise `f` with `x` $\longmapsto [\![$`C(n: 2)`$]\!]$.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` $\longmapsto [\![$`C(n: 2)`$]\!]$

`y` $\longmapsto [\![\bot]\!]$

1. `let y`: extend ctx with `y` $\longmapsto [\![\bot]\!]$

Shape Propagation

Trace 1: `test() = f(C(2))`

Specialise `f` with `x` $\longmapsto [\![$`C(n: 2)`$]\!]$.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` $\longmapsto [\![$`C(n: 2)`$]\!]$

`y` $\longmapsto [\![\bot]\!]$

1. `let y`: extend ctx with `y` $\longmapsto [\![\bot]\!]$
2. `if x is C`: `filter(`$[\![$`C(n: 2)`$]\!]$`, C)` = $[\![$`C(n: 2)`$]\!]$ so `then` is viable; `rest` = $[\![\bot]\!]$ so `else` is dead

Shape Propagation

Trace 1: `test() = f(C(2))`

Specialise `f` with `x` $\longmapsto [\![$`C(n: 2)`$]\!]$.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` $\longmapsto [\![$`C(n: 2)`$]\!]$

`y` $\longmapsto [\![$`2`$]\!]$

1. `let y`: extend ctx with `y` $\longmapsto [\![\bot]\!]$
2. `if x is C`: `filter(`$[\![$`C(n: 2)`$]\!]$`, C)` = $[\![$`C(n: 2)`$]\!]$ so `then` is viable; `rest` = $[\![\bot]\!]$ so `else` is dead
3. In `then`: `sop(x.n) = sel(`$[\![$`C(n: 2)`$]\!]$`, n)` = $[\![$`2`$]\!]$, so `y` $\longmapsto [\![$`2`$]\!]$

Shape Propagation

Trace 1: `test() = f(C(2))`

Specialise `f` with `x` $\longmapsto [\![$`C(n: 2)`$]\!]$.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` $\longmapsto [\![$`C(n: 2)`$]\!]$

`y` $\longmapsto [\![$`2`$]\!]$

1. `let y`: extend ctx with `y` $\longmapsto [\![\bot]\!]$
2. `if x is C`: `filter(`$[\![$`C(n: 2)`$]\!]$`, C)` = $[\![$`C(n: 2)`$]\!]$ so `then` is viable; `rest` = $[\![\bot]\!]$ so `else` is dead
3. In `then`: `sop(x.n) = sel(`$[\![$`C(n: 2)`$]\!]$`, n)` = $[\![$`2`$]\!]$, so `y` $\longmapsto [\![$`2`$]\!]$
4. `y + 1` folds to $[\![$`3`$]\!]$, then `Return 3`

Cached as `f_C_Lit2`; body folds entirely to the constant `3`.

Shape Propagation

Trace 2: `test2(dyn) = f(C(dyn))`

Argument shape: ⟦`C(n: dyn)`⟧.

Specialise `f` with `x` ⟼ ⟦`C(n: dyn)`⟧.

```
1  fun f(x) =                                  mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦`C(n: dyn)`⟧

Shape Propagation

Trace 2: `test2(dyn) = f(C(dyn))`

Argument shape: ⟦`C(n: dyn)`⟧.

Specialise `f` with `x` ⟼ ⟦`C(n: dyn)`⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦`C(n: dyn)`⟧

`y` ⟼ ⟦⊥⟧

1. `let y: y` ⟼ ⟦⊥⟧

Shape Propagation

Trace 2: `test2(dyn) = f(C(dyn))`

Argument shape: ⟦`C(n: dyn)`⟧.

Specialise `f` with `x` ⟼ ⟦`C(n: dyn)`⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦`C(n: dyn)`⟧

`y` ⟼ ⟦⊥⟧

1. `let y`: `y` ⟼ ⟦⊥⟧
2. `if x is C`: `filter(`⟦`C(n: dyn)`⟧`, C)` = ⟦`C(n: dyn)`⟧ so `then` is viable; `rest` = ⟦⊥⟧ so `else` is dead

Shape Propagation

Trace 2: `test2(dyn) = f(C(dyn))`

Argument shape: ⟦`C(n: dyn)`⟧.

Specialise `f` with `x` ⟼ ⟦`C(n: dyn)`⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦`C(n: dyn)`⟧

`y` ⟼ ⟦`dyn`⟧

1. `let y`: `y` ⟼ ⟦⊥⟧
2. `if x is C`: `filter(`⟦`C(n: dyn)`⟧`, C)` = ⟦`C(n: dyn)`⟧ so `then` is viable; `rest` = ⟦⊥⟧ so `else` is dead
3. In `then`: `sop(x.n)` = `sel(`⟦`C(n: dyn)`⟧`, n)` = ⟦`dyn`⟧

Shape Propagation

Trace 2: `test2(dyn) = f(C(dyn))`

Argument shape: ⟦`C(n: dyn)`⟧.

Specialise `f` with `x` ⟼ ⟦`C(n: dyn)`⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦`C(n: dyn)`⟧

`y` ⟼ ⟦`dyn`⟧

1. `let y`: `y` ⟼ ⟦⊥⟧
2. `if x is C`: `filter(`⟦`C(n: dyn)`⟧`, C)` = ⟦`C(n: dyn)`⟧ so `then` is viable; `rest` = ⟦⊥⟧ so `else` is dead
3. In `then`: `sop(x.n)` = `sel(`⟦`C(n: dyn)`⟧`, n)` = ⟦`dyn`⟧
4. `y + 1`: cannot fold (⟦`dyn`⟧ `+ 1`); emit code, result ⟦`dyn`⟧

Cached as `f_C_Dyn`; body keeps the `y = x.n` assignment.

Shape Propagation

Trace 3: `test3() = f(0)`

Argument shape: ⟦0⟧ (literal).

Specialise `f` with `x` ⟼ ⟦0⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦0⟧

Shape Propagation

Trace 3: `test3() = f(0)`

Argument shape: ⟦0⟧ (literal).
Specialise `f` with `x` ⟼ ⟦0⟧.

```
1  fun f(x) =                    mls
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

`x` ⟼ ⟦0⟧

`y` ⟼ ⟦⊥⟧

1. `let y: y` ⟼ ⟦⊥⟧

Shape Propagation

Trace 3: `test3() = f(0)`

Argument shape: $[\![0]\!]$ (literal).
Specialise `f` with $x \mapsto [\![0]\!]$.

```mls
1  fun f(x) =
2    let y
3    if x is C then
4      y = x.n
5    else
6      y = 0
7    y + 1
```

ctx

$x \mapsto [\![0]\!]$

$y \mapsto [\![\bot]\!]$

1. `let y`: $y \mapsto [\![\bot]\!]$
2. `if x is C`: `filter`$([\![0]\!],$ `C)` $= [\![\bot]\!]$ so `then` is dead; `rest`$([\![0]\!],$ `C)` $= [\![0]\!]$ so `else` is viable

Shape Propagation

Trace 3: `test3() = f(0)`

Argument shape: $[\![0]\!]$ (literal).
Specialise `f` with $x \mapsto [\![0]\!]$.

```mls
fun f(x) =
  let y
  if x is C then
    y = x.n
  else
    y = 0
  y + 1
```

ctx

$x \mapsto [\![0]\!]$

$y \mapsto [\![0]\!]$

1. `let y`: $y \mapsto [\![\bot]\!]$
2. `if x is C`: `filter(`$[\![0]\!]$`, C)` = $[\![\bot]\!]$ so `then` is dead; `rest(`$[\![0]\!]$`, C)` = $[\![0]\!]$ so `else` is viable
3. In `else`: $y \mapsto [\![0]\!]$

Shape Propagation

Trace 3: `test3() = f(0)`

Argument shape: ⟦0⟧ (literal).
Specialise `f` with `x` ⟼ ⟦0⟧.

```mls
fun f(x) =
  let y
  if x is C then
    y = x.n
  else
    y = 0
  y + 1
```

ctx

`x` ⟼ ⟦0⟧
`y` ⟼ ⟦0⟧

1. `let y`: `y` ⟼ ⟦⊥⟧
2. `if x is C`: `filter(`⟦0⟧`, C) =` ⟦⊥⟧ so `then` is dead; `rest(`⟦0⟧`, C) =` ⟦0⟧ so `else` is viable
3. In `else`: `y` ⟼ ⟦0⟧
4. `y + 1` folds to ⟦1⟧, then `Return 1`

Cached as `f_Lit0`.

Shape Propagation

Resulting Cache

After all three calls, the staged module contains:

```mls
module Simple withs
  fun f_C_Dyn(x) =          // remove match
    let y
    y = x.n
    y + 1
  fun f_C_Lit2(x) = 3       // fully specialised
  fun f_Lit0()    = 1       // fully specialised
  fun test()      = 3
  fun test2(dyn)  = f_C_Dyn(C(dyn))
  fun test3()     = 1
```

Pattern Matching (Formalization)

Branch Filter (`filter`)

filter removes impossible shapes from the branches of a pattern match.

$$\begin{aligned}
\text{filter}(s, _) &= s & \text{filter}(\llbracket \iota_1 \rrbracket, \iota_2) &= \llbracket \iota_1 \rrbracket \quad \text{if} \quad \iota_1 = \iota_2 \\
\text{filter}(\llbracket [\overline{s_i}^n] \rrbracket, []^n) &= \llbracket [\overline{s_i}^n] \rrbracket & \text{filter}(\llbracket C(\overline{n_i : s_i}^m) \rrbracket, C) &= \llbracket C(\overline{n_i : s_i}^m) \rrbracket \\
\text{filter}(\mathbf{dyn}, \pi) &= \text{silh}(\pi) & \text{filter}(s_1 \cup s_2, \pi) &= \text{filter}(s_1, \pi) \cup \text{filter}(s_2, \pi) \\
\text{filter}(s, \pi) &= \bot \quad \text{otherwise}
\end{aligned}$$

Example

$$\begin{aligned}
\text{filter}(\llbracket C(n : \mathbf{dyn}) \rrbracket, C) &= \llbracket C(n : \mathbf{dyn}) \rrbracket \\
\text{filter}(\llbracket 0 \rrbracket, C) &= \bot \\
\text{filter}(\llbracket C(n : \mathbf{dyn}) \rrbracket \cup \llbracket 0 \rrbracket, C) &= \text{filter}(\llbracket C(n : \mathbf{dyn}) \rrbracket, C) \cup \text{filter}(\llbracket 0 \rrbracket, C) \\
&= \llbracket C(n : \mathbf{dyn}) \rrbracket \cup \bot = \llbracket C(n : \mathbf{dyn}) \rrbracket
\end{aligned}$$

Pattern Matching (Formalization)

Silhouette Function (`silh`)

$\mathrm{silh}(\pi)$ creates the shape of a pattern π, with all sub-shapes set to $\mathbf{dyn}$.

$$\mathrm{silh}(\iota) = [\![\iota]\!] \quad \mathrm{silh}(C) = \left[\!\left[C\left(\overline{\mathbf{dyn}}^{n}\right)\right]\!\right] \quad \mathrm{silh}([]^{n}) = \left[\!\left[\left[\overline{\mathbf{dyn}}^{n}\right]\right]\!\right] \quad \mathrm{silh}(\_) = \mathbf{dyn}$$

Used in $\mathrm{filter}(\mathbf{dyn}, \pi) = \mathrm{silh}(\pi)$: when the scrutinee is $\mathbf{dyn}$, the branch becomes viable with the silhouette shape — a concrete class/literal shell wrapping $\mathbf{dyn}$ sub-shapes.

Pattern Matching (Formalization)

Branch Remainder (`rest`)

rest removes shapes already covered by a match pattern.

$$\begin{aligned}
\mathrm{rest}(s, _) &= \bot & \mathrm{rest}(\llbracket \iota_1 \rrbracket, \iota_2) &= \bot \quad \text{if} \quad \iota_1 = \iota_2 \\
\mathrm{rest}(\llbracket [\overline{s_i}^n] \rrbracket, []^n) &= \bot & \mathrm{rest}(\llbracket C(\overline{n_i : s_i}^m) \rrbracket, C) &= \bot \\
\mathrm{rest}(\mathbf{dyn}, \pi) &= \mathbf{dyn} \quad \text{if} \quad \pi \neq _ & \mathrm{rest}(s_1 \cup s_2, \pi) &= \mathrm{rest}(s_1, \pi) \cup \mathrm{rest}(s_2, \pi) \\
\mathrm{rest}(s, \pi) &= s \quad \text{otherwise}
\end{aligned}$$

Example

$$\begin{aligned}
\mathrm{rest}(\llbracket C(n : \mathbf{dyn}) \rrbracket \cup \llbracket 0 \rrbracket, C) &= \mathrm{rest}(\llbracket C(n : \mathbf{dyn}) \rrbracket, C) \cup \mathrm{rest}(\llbracket 0 \rrbracket, C) \\
&= \bot \cup \llbracket 0 \rrbracket = \llbracket 0 \rrbracket
\end{aligned}$$

After matching the `C` arm, the remainder shape for the `else` branch is $\llbracket 0 \rrbracket$.

Pattern Matching (Formalization)

Non Termination 1

Consider fibonacci number:

```mls
staged module Staged with
  fun fib(n) = if n is
    1 then 1
    2 then 1
    n then fib(n - 1) + fib(n - 2)
```

With `n` $\mapsto$ ⟦`dyn`⟧, every branch is viable. The recursive call `fib(n - 1)` has argument shape ⟦`dyn`⟧, so shape propagation recurses forever, even though the original program terminates for valid inputs.

Pattern Matching (Formalization)

Solution: Stub Insertion

Before propagating into `fib(dyn)`, we **pre-insert a stub** into the cache:

Cache (stub inserted)

Function	Shapes	Symbol	Block	Returned Shape
`fib`	$(\mathbf{dyn},)$	`fib_Dyn`	*stub*	**dyn**

Now when propagating the `n then fib(n - 1) + fib(n - 2)` arm:

1. `fib(n - 1)` has argument shape ⟦`dyn`⟧ — matches the existing stub ⟶ rewritten to `fib_Dyn(n - 1)`.
2. Return shape ⟦`dyn`⟧ is read from the stub; recursion stops.
3. `fib(n - 2)` likewise hits the stub ⟶ rewritten to `fib_Dyn(n - 2)`.
4. Result shape of the arm: ⟦`dyn`⟧ + ⟦`dyn`⟧ = ⟦`dyn`⟧.

Pattern Matching (Formalization)

The completed specialised function will be

```mls
fun fib_Dyn(n) = if n is
  1 then 1
  2 then 1
  n then fib_Dyn(n - 1) + fib_Dyn(n - 2)
```

Cache (stub filled)

Function	Shapes	Symbol	Block	Returned Shape
`fib`	$(\mathbf{dyn},)$	`fib_Dyn`	`if n is ...`	**dyn**

Remark. Stub insertion only guarantees termination of the first stage (shape propagation) when **the original program terminates**.

Pattern Matching (Formalization)

Non Termination 2

Consider a function that scans an array for a zero:

```mls
staged module Staged with
  fun f(xs, i) =
    if xs.(i) is
      0 then i
      v then f(xs, i + 1)
```

With `xs` $\mapsto$ ⟦`dyn`⟧ and `i` $\mapsto$ ⟦`0`⟧ on the first call, `xs.(0)` is ⟦`dyn`⟧ — both branches viable.

The recursive call has `i` $\mapsto$ ⟦`1`⟧, then ⟦`2`⟧, ... — Shape Propagation diverges!

Pattern Matching (Formalization)

Solution: Context Decay

Before processing the branches of `if xs.(i) is`, since the scrutinee has shape ⟦`dyn`⟧, we **decay** the context: every path that influenced the scrutinee is refined to ⟦`dyn`⟧.

Here `xs.(i)` depends on `xs` and `i`, so both are decayed to ⟦`dyn`⟧ before entering the branches.

$$\text{decay}(\Gamma, p, s) = \Gamma \quad \text{if} \;\; p \neq x \text{ for some variable } x \text{ or } s \neq \mathbf{dyn}$$

$$\text{decay}(\Gamma, x, \mathbf{dyn}) = \Gamma \cdot \left[\overline{p_i \mapsto \mathbf{dyn}}\right]^{p_i \in \text{ alias}(\text{clos}(x))}$$

With `i` $\mapsto$ ⟦`dyn`⟧ after decay, `i + 1` $\mapsto$ ⟦`dyn`⟧ too. The recursive call always hits a stable shape — propagation terminates.

Pattern Matching (Formalization)

Combining `filter`, `rest`, dead branch elimination and `decay` yields the following formalization:

$$\frac{\overline{\Gamma_0 \cdot (p \mapsto s_{i'}) \vdash b_i \rightsquigarrow b_{i'} \Rightarrow \Gamma_i, s_{i''}}^{i=1..n, s_i \neq \bot} \qquad \Gamma_0 \bowtie \overline{\Gamma_i - \Gamma_0}^{i=1..n, s_i \neq \bot} \vdash b_r \rightsquigarrow b_{r'} \Rightarrow \Gamma', s \qquad \overline{s_{i'} = \mathrm{filter}(s_{i-1}, \pi_i) \quad s_i = \mathrm{rest}(s_{i-1}, \pi_i)}^{n}}{\Gamma \vdash p \Rightarrow s_0 \qquad \Gamma_0 = \mathrm{decay}(\Gamma, p, s_0)}$$

$$\Gamma \vdash \textbf{if } p \textbf{ is } \overline{\pi_i \textbf{ then } b_i}^{n}; b_r \rightsquigarrow \textbf{if } p \textbf{ is } \overline{\pi_i \textbf{ then } b_{i'}}^{i=1..n, s_i \neq \bot}; b_{r'} \Rightarrow s$$

Staged Class

Similar to staging module, we can stage classes as well, where we specializes on its method, with the main difference being dynamic dispatching `x.f()`.

```
1   staged class B1(val y) with
2     fun call(x) = x + 2 + y
3   staged class B2(val y) with
4     fun call(x) = x + y
5   staged module M with
6     fun twice(f, x) = f.call(f.call(x))
7     fun pick(x, y, b) = if b then x else y
8     fun f(b) =
9       let m = pick(new B1(2), new B2(3), b)
10      twice(m, 5)
```

Staged Class

Trace A: `f(dyn)`

Specialise `f` with `b` $\longmapsto [\![\texttt{dyn}]\!]$.

```
1  fun f(b) =                                    mls
2    let m = pick(new B1(2),new
     B2(3),b)
3    twice(m, 5)
```

f's ctx

`b` $\longmapsto [\![\texttt{dyn}]\!]$

Staged Class

Trace A: `f(dyn)`

Specialise `f` with `b` $\longmapsto [\![\texttt{dyn}]\!]$.

```
1  fun f(b) =                                  mls
2      let m = pick(new B1(2),new B2(3),b)
3      twice(m, 5)
```

f's ctx

`b` $\longmapsto [\![\texttt{dyn}]\!]$

`m` $\longmapsto [\![\bot]\!]$

1. `let m`: extend ctx with `m` $\longmapsto [\![\bot]\!]$

Staged Class

Trace A: `f(dyn)`

Specialise `f` with `b` $\longmapsto [\![\texttt{dyn}]\!]$.

```
1  fun f(b) =                                    mls
2      let m = pick(new B1(2),new B2(3),b)
3      twice(m, 5)
```

f's ctx

`b` $\longmapsto [\![\texttt{dyn}]\!]$

`m` $\longmapsto [\![\bot]\!]$

1. `let m`: extend ctx with `m` $\longmapsto [\![\bot]\!]$
2. Call `pick(new B1(2), new B2(3), b)`: argument shapes are $[\![\texttt{B1(2)}]\!]$, $[\![\texttt{B2(3)}]\!]$, $[\![\texttt{dyn}]\!]$. Recurse into `pick` with a fresh ctx $\rightarrow$ **go to Trace B**

Staged Class

Trace B: `pick(B1(2), B2(3), dyn)`

```
1  fun pick(x, y, b) =                mls
2    if b then x
3    else y
```

pick's ctx

$x \longmapsto [\![\texttt{B1(2)}]\!]$

$y \longmapsto [\![\texttt{B2(3)}]\!]$

$b \longmapsto [\![\texttt{dyn}]\!]$

Staged Class

Trace B: `pick(B1(2), B2(3), dyn)`

```
1  fun pick(x, y, b) =                mls
2    if b then x
3    else y
```

pick's ctx

$x \longmapsto [\![\texttt{B1(2)}]\!]$

$y \longmapsto [\![\texttt{B2(3)}]\!]$

$b \longmapsto [\![\texttt{dyn}]\!]$

1. `if b`: `b` is $[\![\texttt{dyn}]\!]$, so both branches viable

Staged Class

Trace B: `pick(B1(2), B2(3), dyn)`

```
1  fun pick(x, y, b) =          mls
2    if b then x
3    else y
```

pick's ctx

$x \longmapsto [\![\texttt{B1(2)}]\!]$

$y \longmapsto [\![\texttt{B2(3)}]\!]$

$b \longmapsto [\![\texttt{dyn}]\!]$

1. `if b`: `b` is $[\![\texttt{dyn}]\!]$, so both branches viable
2. `then` arm returns `x`, shape = $[\![\texttt{B1(2)}]\!]$

Staged Class

Trace B: `pick(B1(2), B2(3), dyn)`

```
1  fun pick(x, y, b) =          mls
2    if b then x
3    else y
```

`pick`'s ctx

$\texttt{x} \longmapsto [\![\texttt{B1(2)}]\!]$

$\texttt{y} \longmapsto [\![\texttt{B2(3)}]\!]$

$\texttt{b} \longmapsto [\![\texttt{dyn}]\!]$

1. `if b`: `b` is $[\![\texttt{dyn}]\!]$, so both branches viable
2. `then` arm returns `x`, shape = $[\![\texttt{B1(2)}]\!]$
3. `else` arm returns `y`, shape = $[\![\texttt{B2(3)}]\!]$

Staged Class

Trace B: `pick(B1(2), B2(3), dyn)`

```
1  fun pick(x, y, b) =          mls
2    if b then x
3    else y
```

pick's ctx

$x \longmapsto [\![B1(2)]\!]$

$y \longmapsto [\![B2(3)]\!]$

$b \longmapsto [\![dyn]\!]$

1. `if b`: `b` is $[\![dyn]\!]$, so both branches viable
2. `then` arm returns `x`, shape = $[\![B1(2)]\!]$
3. `else` arm returns `y`, shape = $[\![B2(3)]\!]$
4. Result shape: $[\![B1(2) \cup B2(3)]\!]$. Cached as `pick1`. Return to Trace A.

Staged Class

Trace A (continued): back in `f`

```
1  fun f(b) =                                   mls
2    let m = pick(new B1(2), new B2(3), b)
3    twice(m, 5)
```

f's ctx

b ⟼ ⟦dyn⟧

m ⟼ ⟦B1(2) ∪ B2(3)⟧

Staged Class

Trace A (continued): back in `f`

```
1  fun f(b) =                                   mls
2      let m = pick(new B1(2), new B2(3), b)
3      twice(m, 5)
```

`f`'s ctx

$\texttt{b} \longmapsto [\![\texttt{dyn}]\!]$

$\texttt{m} \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

1. Bind `m` to `pick`'s returned shape = $[\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

Staged Class

Trace A (continued): back in `f`

```
1  fun f(b) =                                  mls
2    let m = pick(new B1(2), new
     B2(3), b)
3    twice(m, 5)
```

`f`'s ctx

`b` $\longmapsto$ ⟦`dyn`⟧

`m` $\longmapsto$ ⟦`B1(2)` $\cup$ `B2(3)`⟧

1. Bind `m` to `pick`'s returned shape = ⟦`B1(2)` $\cup$ `B2(3)`⟧
2. `twice(m, 5)`: $\rightarrow$ **go to Trace C** and specialise as `twice1`

Staged Class

Trace C: `twice({B1(2), B2(3)}, 5)`

```mls
fun twice(f, x) =
  let tmp
  tmp = f.call(x)
  f.call(tmp)
```

twice's ctx

$f \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

$x \longmapsto [\![5]\!]$

Recall:

```mls
class B1(y) with
  fun call(x) = x + 2 + y
class B2(y) with
  fun call(x) = x + y
```

Staged Class

Trace C: `twice({B1(2), B2(3)}, 5)`

```mls
fun twice(f, x) =
  let tmp
  tmp = f.call(x)
  f.call(tmp)
```

Recall:

```mls
class B1(y) with
  fun call(x) = x + 2 + y
class B2(y) with
  fun call(x) = x + y
```

twice's ctx

$\texttt{f} \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

$\texttt{x} \longmapsto [\![\texttt{5}]\!]$

$\texttt{tmp} \longmapsto [\![\bot]\!]$

1. `let tmp`: extend ctx with $\texttt{tmp} \longmapsto [\![\bot]\!]$

Staged Class

Trace C: `twice({B1(2), B2(3)}, 5)`

```mls
fun twice(f, x) =
  let tmp
  tmp = f.call(x)
  f.call(tmp)
```

Recall:

```mls
class B1(y) with
  fun call(x) = x + 2 + y
class B2(y) with
  fun call(x) = x + y
```

twice's ctx

$\texttt{f} \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

$\texttt{x} \longmapsto [\![\texttt{5}]\!]$

$\texttt{tmp} \longmapsto [\![\bot]\!]$

2. `f.call(x)` on union → **split by class**:
   - `f` = $[\![\texttt{B1(2)}]\!]$:

     `B1.call`'s ctx

     $\texttt{this} \longmapsto [\![\texttt{B1(2)}]\!]$

     $\texttt{x} \longmapsto [\![\texttt{5}]\!]$

     cache `f.call1()` in `B1` → $[\![\texttt{9}]\!]$
   - `f` = $[\![\texttt{B2(3)}]\!]$: cache `f.call1()` in `B2` → $[\![\texttt{8}]\!]$

Staged Class

Trace C: `twice({B1(2), B2(3)}, 5)`

```mls
1  fun twice(f, x) =
2    let tmp
3    tmp = f.call(x)
4    f.call(tmp)
```

Recall:

```mls
1  class B1(y) with
2    fun call(x) = x + 2 + y
3  class B2(y) with
4    fun call(x) = x + y
```

twice's ctx

$\texttt{f} \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

$\texttt{x} \longmapsto [\![\texttt{5}]\!]$

$\texttt{tmp} \longmapsto [\![\texttt{9} \cup \texttt{8}]\!]$

3. `tmp = f.call(x)`: bind `tmp` to $[\![\texttt{9} \cup \texttt{8}]\!]$

Staged Class

Trace C (continued): second call

```mls
1  fun twice(f, x) =
2    let tmp
3    tmp = f.call(x)
4    f.call(tmp)
```

Recall:

```mls
1  class B1(y) with fun call(x) = x + 2 + y
2  class B2(y) with fun call(x) = x + y
```

twice's ctx

$\texttt{f} \longmapsto [\![\texttt{B1(2)} \cup \texttt{B2(3)}]\!]$

$\texttt{x} \longmapsto [\![\texttt{5}]\!]$

$\texttt{tmp} \longmapsto [\![\texttt{9} \cup \texttt{8}]\!]$

4. `f.call(tmp)` with `tmp` in $[\![\texttt{9} \cup \texttt{8}]\!]$: split again, body kept as code
   - `f` = $[\![\texttt{B1}]\!]$: cache `f.call2(x) = x + 2 + 2` in `B1` $\longrightarrow [\![\texttt{13} \cup \texttt{12}]\!]$
   - `f` = $[\![\texttt{B2}]\!]$: cache `f.call2(x) = x + 3` in `B2` $\longrightarrow [\![\texttt{12} \cup \texttt{11}]\!]$

Staged Class

Resulting Cache

After propagation, the residual `M` and the residual classes contain:

```mls
module M with
  fun f(b) =
    let m, tmp1, tmp2
    tmp1 = new B1(2)
    tmp2 = new B2(3)
    m = pick1(tmp1, tmp2, b)
    twice1(m)
  fun pick1(x, y, b) =
    if b then new B1(2) else new B2(3)
```

```mls
  fun twice1(f) =
    let tmp
    if f is
      B1 then tmp = f.call1()
      B2 then tmp = f.call1()
    if f is
      B1 then f.call2(tmp)
      B2 then f.call2(tmp)
```

Staged Class

```mls
class B1(y) with
  fun call1() = 9
  fun call2(x) =
    let tmp
    tmp = x + 2
    tmp + 2

class B2(y) with
  fun call1() = 8
  fun call2(x) = x + 3
```

Remark. If `f`'s shape included an **unstaged** class `B3`, we could not specialise the `B3.call` function without its Staged Block. The generated `twice1` would then include an `else` branch falling back to the original `f.call(...)`.

Inlining

A typical module after Dynamic Staging will look like the following.

Dynamic Staging Output

```mls
module M with
  fun pow_Dyn_Lit1(x) =
    let {tmp, tmp1}
    x
  fun pow_Dyn_Lit2(x) =
    let {tmp, tmp1}
    tmp = 1
    tmp1 = pow_Dyn_Lit1(x)
    *(x, tmp1)
```

```mls
  fun pow_Dyn_Lit3(x) =
    let {tmp, tmp1}
    tmp = 2
    tmp1 = pow_Dyn_Lit2(x)
    *(x, tmp1)
```

Inlining

During **the second stage of compilation**, we utilize the MLscript compiler's existing inlining capabilities to inline function calls, as recursion has been eliminated during dynamic staging.

```JavaScript
let x, inlinedVal, tmp1, x1, inlinedVal1, tmp11, x2, inlinedVal2;
x = 2;
x1 = x;
x2 = x1;
inlinedVal2 = x2;
tmp11 = inlinedVal2;
inlinedVal1 = x1 * tmp11;
tmp1 = inlinedVal1;
inlinedVal = x * tmp1;
return inlinedVal
```

Printing Staged Block

After all the specialized functions are completed, we need to write the functions to a new file.

```mls
staged module M with
  val funCache = new Map([
    ["f1", ...],
    ["f2", ...],
    ["g1", ...]
  ])
```

```mls
module M with
  fun f1(x, y) = ...
  fun f2() = ...
  fun g1() = ...
```

Printing Staged Block

This is a similar process to staging, but done in reverse.

As before, we need extra care when handling symbols.

Model-View-Projection transformation

basically we're pretty good at partially evaluating matrices ^w^.

Time: about 2x (from 2s to 1s, eh...)

Code size: don't worry about it, we'll just do lifting

Bibliography

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